

Stefan-Razvan Balauca

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EDUCATION

PhD in Computer Science – Quantum Computing and Machine Learning

Alexandru Ioan Cuza University

Iasi, Romania

Oct. 2025 – present

- Research Topic: Towards Robust Quantum Machine Learning - [Research Proposal](#)

Research Master – Advanced Studies in Computer Science

Alexandru Ioan Cuza University

Iasi, Romania

Oct. 2021 – July 2023

- Average Grade: 10 - 1st place. Thesis Grade: 10
- Thesis: Links between Batch Correction, Covariation and Gene Regulatory Networks - [Thesis](#)
- Research Topics: Quantum Computing, Machine Learning, Deep Learning, NLP, Security

Bachelor of Science in Computer Science

Alexandru Ioan Cuza University

Iasi, Romania

Oct. 2018 – July 2021

- Average Grade: 9.97 - 1st place. Thesis Grade: 10
- Thesis: An analysis of Quantum Circuits for simulating Multi-Controlled Toffoli Gates - [Thesis](#), [Abstract](#)
- Main interests: Neural Networks, Machine Learning, Quantum Computing, Genetic Algorithms, Cryptography

Bachelor of Engineering in Technological Physics

Alexandru Ioan Cuza University

Iasi, Romania

Oct. 2018 – July 2022

- Average Grade: 10 - 1st place. Thesis Grade: 10
- Thesis: The Technology of Self-Organising Robots: Challenges and Applications - [Thesis \(Romanian\)](#)
- Main interests: Quantum Mechanics, Statistical Physics, Physics of Plasma, Lasers, Nuclear Physics, Astrophysics, Electronics, Electrodynamics

EXPERIENCE

Assistant Lecturer – Quantum Computing, ML and Numerical Calculus

Alexandru Ioan Cuza University

Oct. 2026 - Present

Iasi, Romania

- Design and evaluate homework and projects; Explain lab exercises to BSc students; Supervise BSc and MSc theses

Researcher – Quantum Computing and Machine Learning

FreeYaMind Campus

Jan. 2025 - Present

Iasi, Romania

- Research Topics: Near-term Applications of Quantum Computing and Machine Learning

Researcher – Secure and Reliable AI

INSAIT, Sofia University “St. Kliment Ohridski”

Sept. 2023 - Jan. 2025

Sofia, Bulgaria

- Research Topic: Certified Robustness of Neural Networks, Adviser: [Prof. Martin Vechev](#) (ETH Zürich)

Research Intern

STEM Cell Institute - University of Cambridge

May 2023 – Aug. 2023

Cambridge, UK

- Exploring Gene Expression covariation and designing algorithms for Gene Regulatory Network inference

Teaching Assistant – Quantum Computing, Numerical Calculus and ML

Alexandru Ioan Cuza University

Oct. 2021 – July 2023

Iasi, Romania

- Design and evaluate homework and projects; Explain exercises and problems to BSc students

Research Intern

University of Maryland

July 2021 – Oct. 2021

College Park, MD, USA

- Characterizing, Quantifying and Remediating Label Noise in Malware Detection using Machine Learning

Software Developer – Mathematics Specialist Intern

Gemini CAD Systems

July 2020 – Sep. 2020

Iasi, Romania

- Developed a Photometric Stereo algorithm for processing multiple images of materials into generating texture maps (normal map and height map)
- Sketched the algorithm in Python, ported it to C++ and optimised it to use graphical processing (CUDA)

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, C#, Assembly, SQL, JavaScript, HTML/CSS, MatLab, R
Developer Tools: Git, Docker, Google Cloud Platform, Azure, VS Code, Visual Studio, IntelliJ, DataGrip, Unity
Languages: Romanian (native), English (C1 - TOEFL iBT), French(High School level)

SCIENTIFIC PUBLICATIONS

International Conferences and Journals

- *Experimental Comparison of a Classical CNN and a Hybrid Quantum-Classical Model for Dermoscopic Skin Lesion Classification* - A.-G. Panca, M. Cioratanu, **S. Balauca**, A. Iftene - 30th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (**KES 2026**, Dublin, Ireland)
- *Practical Hybrid Quantum Language Models with Observable Readout on Real Hardware* - **S. Balauca**, A.-A. Balauca, A. Iftene (**arXiv preprint, 2025** – under submission at NeurIPS 2026; **paper**)
- *Gaussian Loss Smoothing Enables Certified Training with Tight Convex Relaxations* - **S. Balauca**, M.N. Müller, Y. Mao, M. Baader, M. Fischer, M. Vechev - Transactions on Machine Learning Research (**TMLR 2025**; **paper**)
- *CTBench: A Library and Benchmark for Certified Training* - Y. Mao, **S. Balauca**, M. Vechev - 42nd International Conference on Machine Learning (**ICML 2025**, Vancouver, Canada; **paper**)
- *MixAT: Combining Continuous and Discrete Adversarial Training for LLMs* - C. Dékány, **S. Balauca**, R. Staab, D.I. Dimitrov, M. Vechev - 39th Conference on Neural Information Processing Systems (**NeurIPS 2025**; **paper**)
- *Understanding Museum Exhibits using Vision-Language Reasoning* - A.-A. Balauca, S. Garai, **S. Balauca**, et al. - IEEE/CVF International Conference on Computer Vision (**ICCV 2025**, Hawaii, US; **paper**)
- *Efficient Constructions for Simulating Multi Controlled Quantum Gates* - **S. Balauca**, A. Arusoae - 22nd International Conference on Computational Science (**ICCS 2022**, London, UK; **paper**, oral, co-chair)
- *Finite difference method for modeling the dielectric properties of ceramic composites* - **S. Balauca**, L. Padurariu, L. Mitoseriu - 13th Conference for Young Scientist in Ceramics, (**CYSC 2019**, Novi-Sad, Serbia; **oral** + **slides**)

Other Conferences

- *Study of forced oscillations using the fazorial method* - **S. Balauca**, S. Popescu - 57th National Physics and Modern Educational Technologies Conference, May 2019, Iasi, Romania (**oral** + **slides**)

RESEARCH PROJECTS

- Visual Word Sense Disambiguation - SemEval 2023** | *pytorch, nltk, wordnet* Oct 2022 – Feb 2023
- Convolutional, recurrent and attention networks for identifying the best image describing an ambiguous word
- Summarizing a story** | *nltk, SpaCy, Google Colab* Oct 2020 – Jan 2021
- Extracting the main characters from a story and correlating them with their explicit and implicit actions
 - Winner of the Amazon Learn and Earn competition
- Q-learning algorithm for gym's CarRacing environment** | *keras, tensorflow* Oct 2020 – Jan 2021
- Obtained scores of 850-900/1000, better than human performance
- Participation at CMMRS Research Summer School** | *MPI-SWS, Saarbrücken, DE* Aug 2022
- Attended lectures on cutting-edge research topics in computer science, including theory of deep learning, large-scale machine learning, NLP, distributed systems, security and privacy

AWARDS

International Awards - Mathematics

- **Silver Medal** - 28th International Mathematics Competition for University Students, 2021, on-line
- **Gold Medal** - 13th International Ariel Mathematical Olympiad, 2021, on-line
- **Gold Medal** - 12th International Ariel Mathematical Olympiad, 2020, on-line
- **Gold Medal** - 14th South-Eastern Europe Mathematical Olympiad for University Students, 2020, Thessaloniki, Gr
- **Gold Medal** - 11th International Ariel Mathematical Olympiad, 2019, Ariel, Israel
- **Gold Medal - Team Contest** - 11th Ariel Mathematical Olympiad, 2019, Ariel, Israel
- **Silver Medal** - 13th South-Eastern Europe Mathematical Olympiad for University Students, 2019, Devin, Bulgaria
- **Silver Medal** - 19th Balkan Mathematical Olympiad, 2015, Belgrade, Serbia

Artificial Intelligence and Programming

- **1st place worldwide** - 36th Catalyst CloudFlight Coding Contest - AI, Nov 2022, online
- **2nd place Iasi, 19th place worldwide** - 35th Catalyst CloudFlight Coding Contest - AI, Nov 2021, online
- **15th place Romania, 265th place worldwide** - Google HashCode Contest, Feb. 2020, online
- **2nd place Iasi, 7th place Romania** - ACM ICPC Subregional Contest, December 2020, Iasi, Romania
- **Best Score** - Amazon's "Learn & Earn" Competition within "Data Structures" discipline, Feb 2019
- **Best Project** - Amazon's "Learn & Earn" Competition within "Artificial Intelligence" discipline, Jan 2021

International Awards - Physics and Sciences

- **Gold Medal** - 19th Asiatic Physics Olympiad, 2018, Hanoi, Vietnam
- **Gold Medal** - 11th International Earth Sciences Olympiad, 2017, Nice, France
- **Silver Medal** - 12th International Junior Sciences Olympiad, 2015, Daegu, South Korea
- **Silver Medal** - 11th International Junior Sciences Olympiad, 2014, Mendoza, Argentina

International Awards - Linguistics

- **Bronze Medal** - 15th International Linguistics Olympiad, 2017, Dublin, Ireland
- **Mention - Team Contest** - 15th International Linguistics Olympiad, 2017, Dublin, Ireland

Other Awards

- **8 Gold Medals and prizes (top 3)** - National Mathematics, Physics, Linguistics and Sciences Olympiads in High School (2014 - 2018)